Simulate a circuit and practice soldering

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test a circuit representation before making permanent connections.

SUCCESS CRITERIA - I can...

1. Build a complete battery-switch-motor circuit in the simulation.
2. Explain an open versus closed path.
3. Document a supervised solder-practice check.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “open circuit” and “short circuit” to explain your answer. Sentence starters are optional.

open circuit: a break that prevents current through that path

short circuit: an unintended low-resistance path bypassing a component

insulation: material that limits unwanted electric conduction

01 Warm-up: bring back an earlier idea

A car receives **50 J** electrical input but gains only **30 J** translational kinetic energy. **Name two possible destinations of** the difference and **explain why the account needs** them.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $I = \Delta q/\Delta t$; $I = 0$ in an ideal open circuit.
Use the assigned simulation and teacher solder procedure. Disconnect power; hot irons/joints stay in the designated work area.

Worked reasoning: In a simple car circuit, battery positive connects through the switch and motor back to battery negative. A wire directly across the battery bypasses the motor and can create a hazardous short circuit.

Your turn: Draw the intended closed loop and circle a place where an open switch safely breaks it.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Draw your simulated closed circuit** and an open-switch version.
2. **Predict which runs the motor** and why.

Sentence starter: I predict … . I used … to make this prediction because … .



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: 1 2 3

04 Apply the physics

A test resistor has **6 V** across it and carries **0.30 A**.
1. **Calculate resistance** and power.
2. **Compare it** with the motor circuit shown: why might a running motor fail to behave like that fixed resistor?

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

05 Explain your reasoning

Make a two-column record: practice-joint observation and next improvement. Include teacher feedback on the joint and tool handling.

Sentence starter: I notice … . This supports … because … .

Observed joint / handling	Teacher feedback	Next time
___	___	___
___	___	___

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Two equal **12 Ω** resistors are connected in parallel. **Find equivalent resistance**, then compare with a series connection of the same pair.
Draw the stated circuit and **label quantities relevant** to the question.

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Explain why a shiny joint alone does not show that the electrical and mechanical connection is reliable.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **6 V** motor circuit is modeled with **3 ohms** while running and **1 ohm** at stall. **Compare currents** and input powers; **explain why resistance**/current may change with motion and what this calculation leaves out.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3